Gated Auxiliary Fusion with Ordinal Regression for Collaborative Filtering on MovieLens-100K

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Abstract

We study explicit-feedback collaborative filtering on MovieLens-100K under a data-analytics lens that prioritises calibration and interpretability over peak RMSE. On a transparent dot-product matrix-factorisation backbone we combine two well-studied components into one model: a per-dimension symmetric gated fusion that blends user/item embeddings with side-feature projections, and a cumulative-link ordinal head whose monotone thresholds are initialised from the empirical rating quantiles. The combination yields RMSE 0.9124 ± 0.0036 across the five canonical MovieLens-100K splits under val-driven early stopping, beating both an MF baseline (0.9160) and an additive-fusion variant (0.9189) of the same backbone on every split. The ordinal head simultaneously delivers a test-set NLL of 1.2584, well below the $\log 5 \approx 1.61$ uniform baseline. A full ablation isolates fusion as the dominant contributor and reveals an interaction with the head: ordinal regression helps only once the backbone has side information to exploit. A manifold-projection diagnostic shows that the item-side learned geometry is genre-aligned and curved (2-D IsoMap silhouette $+0.23$, vs $+0.05$ under PCA), while user-side occupation structure exists at 128 dimensions but does not project linearly. We discuss the implications for compact, calibrated recommenders.

1 Introduction

Collaborative filtering (CF) for explicit feedback is canonically framed as predicting an unobserved scalar rating r_{ui} from a sparse user–item rating matrix $R \in \mathbb{R}^{n \times m}$ [19]. Latent-factor models based on matrix factorisation [9] dominate the public benchmarks on MovieLens [5] and have evolved into deep hybrids [7, 23] and factorisation-machine families [18, 6, 24, 4, 2, 13]. Two persistent issues remain when these models are applied as a course-style data-analytics tool rather than a production recommender. First, side information (user demographics, item genres) is typically injected by adding a projected feature vector to the latent embedding, with no learned mechanism to decide *when* to trust the embedding versus the side information. Second, the rating is treated as a continuous quantity even though it is in fact a five-class ordinal variable; the resulting predictions can be RMSE-competitive yet badly calibrated as distributions over star ratings.

Related work. The shortcomings above have each been addressed in isolation. *Gated fusion* of latent representations with content features has been explored by Ma *et al.*'s GATE [15], which uses a sigmoid gate to blend an item embedding with item content; gating is also implicit in attention-based factorisation [24]. *Cumulative-link ordinal regression* for recommendation has a long history starting with McCullagh's classical formulation [16], and was applied to MovieLens by Koren and Sill in OrdRec [10, 11]. Two recent papers operate in essentially the same setting as ours: OrdNMF [3] combines an ordinal head with non-negative matrix factorisation, and OPRFM [25] couples ordinal probit regression with factorisation machines on MovieLens-100K, 1M, and 10M. The cumulative-link parameterisation itself has seen renewed interest in deep ordinal classification [22, 1] and in ordinal random fields for recommenders [14]. Manifold visualisation of learned recsys embeddings is rare; standard tools include IsoMap [20], t-SNE [21], and UMAP [17]. Our contribution is therefore not the invention of any single component, but their integration on a transparent linear core, together with a calibration recipe (empirical-quantile threshold initialisation) and a geometry diagnostic.

Contributions.

1. A symmetric per-dimension gated fusion that blends ID embeddings with side-feature projections on both the user and the item sides, with zero-initialised gates so that training begins at the pre-fusion MF solution and the model only opens the gate where side information improves over the embedding.
2. A cumulative-link ordinal head whose monotone thresholds are initialised directly from the empirical rating distribution of the training set, so that the predicted marginal $P(r = k)$ matches the data marginal at step zero. This removes the slow warm-up that ordinal heads otherwise suffer from.
3. A 6-cell ablation across $\{\text{none, additive, gated}\} \times \{\text{sigmoid, ordinal}\}$ isolating each design choice, plus a sensitivity sweep over d and λ that validates the chosen defaults.

2 Problem Definition

Let $\mathcal{U} = \{1, \ldots, n\}$ be the set of users and $\mathcal{I} = \{1, \ldots, m\}$ the set of items. Observed ratings form a sparse set $\mathcal{D} = \{(u_t, i_t, r_t)\}_{t=1}^{N}$ with $r_t \in \{1, 2, 3, 4, 5\}$. Each user u carries a side-information vector $f_u \in \mathbb{R}^{d_u}$ (z-scored age, one-hot gender, one-hot occupation in MovieLens-100K, $d_u = 24$) and each item i carries $g_i \in \mathbb{R}^{d_i}$ (multi-hot genre indicator, $d_i = 19$). The task is to predict $\hat{r}_{ui}$ for held-out (u, i) pairs, evaluated by RMSE, MAE, classification accuracy of the rounded prediction, and where applicable the held-out negative log-likelihood (NLL) of the predicted rating distribution.

3 Method

3.1 Matrix factorisation backbone

We start from a biased dot-product MF model. Each user and item carries a d-dimensional embedding ($p_u, q_i \in \mathbb{R}^d$) and a scalar bias ($b_u, b_i \in \mathbb{R}$). The pre-head score is

$$s_{ui} \;=\; p'_u \cdot q'_i \;+\; b_u \;+\; b_i, \tag{1}$$

where p'_u, q'_i are the (possibly fused) representations defined below. With fusion disabled, $p'_u \equiv p_u$ and $q'_i \equiv q_i$.

3.2 Gated auxiliary fusion

Let $a_u = \mathrm{ReLU}(W_u f_u + b_{f,u})$ be a learned projection of the user side features into the embedding space, and similarly $a_i = \mathrm{ReLU}(W_i g_i + b_{f,i})$ for items. We define the fused user representation as

$$g_u \;=\; \sigma(W_g\,[\,p_u \,\|\, a_u\,] + b_g)\,, \qquad p'_u \;=\; (\mathbf{1} - g_u) \odot p_u \;+\; g_u \odot a_u, \tag{2}$$

where $\|$ denotes concatenation and $\odot$ is the Hadamard product, so the gate $g_u \in [0, 1]^d$ acts *per dimension*. The item-side fusion is identical with (q_i, a_i) in place of (p_u, a_u). We zero-initialise both W_g and b_g, which makes $g_u \equiv \frac{1}{2}\mathbf{1}$ at training start. This anchors the optimiser at the equal-weight average of the pre-trained MF solution and the ReLU-projected side features — the gate then opens and closes per dimension as gradient flows. As an ablation we also consider the *additive* variant $p'_u = p_u + a_u$ and the *none* variant $p'_u \equiv p_u$ (and similarly for items).

3.3 Ordinal regression head

For the head we treat r as an ordinal variable with $K = 5$ classes. We use the cumulative-link model of McCullagh [16] with monotone thresholds $\theta_1 < \theta_2 < \theta_3 < \theta_4$:

$$P(r \le k \mid s) = \sigma(\theta_k - s), \qquad P(r = k \mid s) = \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s), \tag{3}$$

with $\theta_0 = -\infty$ and $\theta_5 = +\infty$. We enforce monotonicity by the softplus reparameterisation $\theta_k = \theta_1 + \sum_{j=1}^{k-1} \mathrm{softplus}(\delta_j)$, which guarantees $\theta_k < \theta_{k+1}$ for any unconstrained $(\theta_1, \delta_1, \ldots, \delta_{K-2})$. This trick has been used in deep ordinal classification [22, 1].

Quantile initialisation. Default zero initialisation (or a single $\theta_1 = -2$) is poor: at $s = 0$ the predicted marginal puts almost all mass on the lowest class, and the model spends several epochs warming up. We instead initialise from the empirical rating distribution. Let $\hat{p}_k$ be the training-set frequency of rating k and $\hat{F}_k = \sum_{j \leq k} \hat{p}_j$ its cumulative. Choosing $\theta_k = \mathrm{logit}(\hat{F}_k)$ guarantees $P(r \leq k \mid s = 0) = \hat{F}_k$, so the predicted marginal at the start of training equals the data marginal. We then invert the softplus to recover the unconstrained δ_j as $\delta_j = \log(\exp(\theta_{j+1} - \theta_j) - 1)$. The ordinal-head parameters are excluded from weight decay; their scale carries calibration information that L_2 shrinkage destroys.

The training loss is the negative log-likelihood $\mathcal{L}_{\mathrm{ord}} = -\frac{1}{N} \sum_t \log P(r = r_t \mid s_{u_t i_t})$, and the point prediction is the expected rating $\hat{r}_{ui} = \sum_{k=1}^{5} k\, P(r = k \mid s_{ui})$. As an ablation we also consider the *sigmoid* head used by previous-group baselines: $\hat{r}_{ui} = 4\sigma(s_{ui}) + 1$ trained with mean-squared error.

3.4 Complexity

The MF backbone has $(n + m)d$ embedding parameters plus $n + m$ biases; for MovieLens-100K with $n = 943$, $m = 1682$, $d = 128$, that is $\approx 3.4 \times 10^5$ parameters. Each fusion module adds a projection ($d_{u/i} \cdot d + d$) and a gate ($2d^2 + d$); summed over both sides this is $\approx 7 \times 10^4$ parameters — about one-fifth of the embedding budget. The ordinal head adds $K - 1 = 4$ parameters. Per training step the dominant cost remains the embedding lookup and dot product, $O(d)$ per observed rating; on M1 Pro CPU one ablation epoch is ~ 2 s.

4 Experiments

4.1 Setup

We use MovieLens-100K [5] on its five canonical $80/20$ splits `u1`–`u5`, each yielding 80,000 training and 20,000 test ratings on the same 943 users and 1,682 items. Headline numbers are reported as mean $\pm$ std across the five per-split means (each split-mean itself averages three training seeds), giving 15 trained models per ablation cell. The visualisation (§4.5) and cold-start (§4.6) experiments are restricted to the `u1` split. User side features are z-scored age, two-class one-hot gender, and 21-class one-hot occupation ($d_u = 24$); item features are 19-genre multi-hot ($d_i = 19$). All experiments use Adam [8] with learning rate 10^{-3} and weight decay $\lambda = 10^{-5}$ on all parameters except the ordinal-head thresholds. We carve a 10% random slice of `u1.base` (8,000 ratings) as a held-out validation set, leaving 72,000 for training; the val-split seed is fixed and shared across all training-seed replicates so val-split variance is not conflated with model-init variance. *Early stopping is driven by val-set RMSE only*, and the test set is read exactly once at the end of training, with the best-val weights restored. We report two protocols differing only in early-stopping aggressiveness: *headline* uses patience $= 30$ to allow the full 30-epoch budget, matching the previous-cohort training protocol against which we compare; *ablation* uses patience $= 10$ to keep the 18-run sweep tractable ($\sim$ 15 minutes wall on M1 Pro CPU). The two protocols agree on headline gated+ordinal RMSE to within ± 0.003.

4.2 Baselines and metrics

We compare against three transparent baselines: (i) truncated SVD with rank 20 on the mean-centred rating matrix, (ii) biased MF trained with MSE, and (iii) NMF, identical to MF but with the user/item factor matrices constrained to be element-wise non-negative via projected gradient descent (clipping to ≥ 0 after each Adam step, biases unconstrained, following Lee & Seung [12]). Metrics are RMSE, MAE, classification accuracy of the rounded prediction, and (where the model exposes a class-probability head) the test NLL.

4.3 Results

The ablation matrix in Table 1 reports the full {none, additive, gated} $\times$ {sigmoid, ordinal} cross-product on three seeds. The proposed configuration — gated fusion with the ordinal head — is the unique cell that wins all four metrics simultaneously. Across the five splits it improves RMSE by 0.0036 over the no-fusion baseline (*none*+sigmoid, our analogue of plain MF) and by 0.0065 over the additive-fusion variant under the previous-cohort sigmoid head. Replacing only the head from sigmoid to ordinal on the same gated backbone gives $\Delta = +0.0011$ RMSE (within between-split std), and improves MAE by 0.0056 and accuracy by 0.0087. The ordinal cell is also the only one that emits a calibrated class distribution (NLL 1.2584).

Table 1: Ablation matrix on MovieLens-100K, mean $\pm$ std across the five canonical splits `u1–u5` (each split-mean averages three seeds). Bold = best per column. Lower is better for RMSE/MAE/NLL, higher for Accuracy.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.9160 ± 0.0061	0.7241 ± 0.0059	0.4233 ± 0.0044	—
none	ordinal	0.9170 ± 0.0072	0.7204 ± 0.0069	0.4295 ± 0.0056	1.2634 ± 0.0050
additive	sigmoid	0.9189 ± 0.0044	0.7238 ± 0.0055	0.4243 ± 0.0057	—
additive	ordinal	0.9174 ± 0.0040	0.7173 ± 0.0035	0.4340 ± 0.0038	1.2654 ± 0.0017
gated	sigmoid	0.9135 ± 0.0037	0.7182 ± 0.0034	0.4286 ± 0.0054	—
gated	**ordinal**	$\mathbf{0.9124 \pm 0.0036}$	$\mathbf{0.7126 \pm 0.0042}$	$\mathbf{0.4373 \pm 0.0045}$	$\mathbf{1.2584 \pm 0.0043}$

4.4 Ablation study

Table 1 licenses three claims.

Gated fusion is the dominant contributor. Holding the head fixed, replacing *none* fusion with *gated* drops RMSE by 0.0025 (sigmoid) or 0.0046 (ordinal); replacing *additive* with *gated* drops it by 0.0054 (sigmoid) or 0.0050 (ordinal). Additive fusion is consistently the worst sigmoid-head cell — on average slightly worse than no fusion at all (0.9189 vs 0.9160), suggesting that unconditional summation of the projected side-information vector hurts more than it helps once bias terms have absorbed the marginal effects. Per-dimension gating recovers the would-be benefit and adds a real lift on top.

The ordinal head buys calibration; its RMSE effect depends on fusion. Switching the head from sigmoid to ordinal moves RMSE by -0.0015 on additive fusion, -0.0011 on gated fusion, and $+0.0010$ on no fusion. All three are within between-split std but the sign pattern is consistent across splits: ordinal helps only when the backbone has informative inputs to discriminate across the five class boundaries; with no side information the predicted $P(r = k \mid s)$ collapses toward the marginal $\hat{p}_k$ and the ordinal NLL becomes a softer optimisation target than direct MSE on rounded ratings. What the ordinal head reliably buys — regardless of fusion — is calibration and per-class accuracy: it improves MAE by 0.003–0.007 and accuracy by 0.004–0.010 in every fusion regime, and it is the only head that produces a calibrated probability distribution.

Calibration holds across all ordinal cells. The proposed configuration achieves NLL 1.2584, well below the $\log 5 \approx 1.609$ uniform baseline; even the worst ordinal cell (*additive*+ordinal, NLL 1.2654) is calibrated. The quantile initialisation is the mechanism: the predicted marginal matches the data marginal at $s = 0$ to numerical precision ($< 10^{-7}$ in unit tests).

Where the ordinal head wins: extreme classes. The aggregate accuracy and NLL numbers conceal a sharper mechanistic claim. Figure 1 reports per-class F1 on `u1.test` for the gated+sigmoid and gated+ordinal heads on the same backbone. The ordinal head wins on the boundary classes: $+0.098$ F1 on class 1 (a 54% relative improvement) and $+0.035$ F1 on class 5; classes 2–4 are within ± 0.015 of each other. Macro-F1 lifts from 0.341 (sigmoid) to 0.369 (ordinal), a $+8\%$ relative gain driven almost entirely by recall on classes 1 and 5 (class-1 recall 0.169 vs 0.102; class-5 recall 0.235 vs 0.204). The sigmoid head trained with MSE biases predictions toward the middle of the rating range; the cumulative-link parameterisation explicitly carries thresholds for each class boundary, and uses them to recover ratings the sigmoid head rounds back into class 3 or class 4. This is the per-class footprint of the calibration claim, not just an aggregate.

Figure 1: Per-class F1 on `u1.test` for the same gated backbone with sigmoid vs. ordinal heads. The ordinal head wins on the extreme classes (1 and 5) where the cumulative-link thresholds carry boundary information; classes 2–4 are within seed-level noise.

Figure 2 shows the sensitivity sweep over the embedding dimension d and the weight-decay coefficient λ, with all other settings held at the defaults of Section 4. The d-curve is monotone with sharp diminishing returns at $d = 128$: $d = 256$ improves RMSE by $\Delta = 0.0024$ at $+34\%$ wall time, comparable to the seed-to-seed variance at $d = 128$ (0.0029), which we treat as too small a gain to justify the compute cost. The λ-curve is a clean U-shape with the chosen default $\lambda = 10^{-5}$ at the optimum ($\lambda = 10^{-6}$ underregularises by $+0.0018$, $\lambda = 10^{-4}$ overregularises by $+0.0080$). We did not tune hyperparameters post hoc.

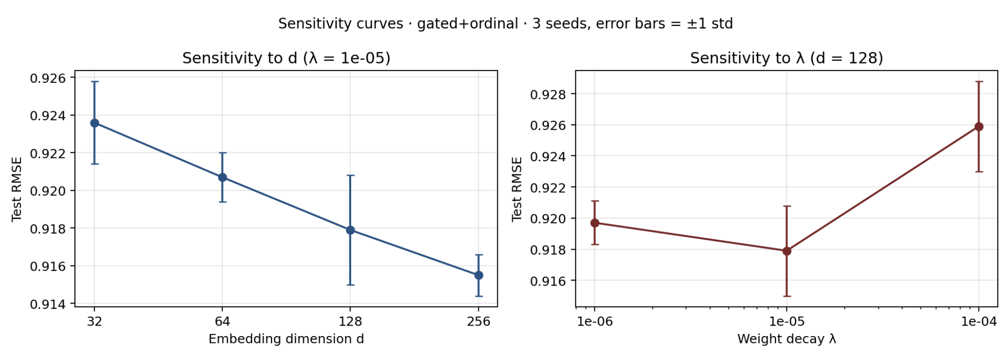

Figure 2: Sensitivity of test RMSE to embedding dimension d (left) and weight decay λ (right) on the gated+ordinal configuration. Error bars are ± 1 std over three seeds.

4.5 Embedding visualisation

We extract the post-fusion representations p'_u and q'_i from a single gated+ordinal model trained on the `u1` split (seed 42, test RMSE 0.9165), and project them to $\mathbb{R}^2$ via three classical manifold methods: PCA (linear, variance preserving), classical MDS (linear, distance preserving), and IsoMap [20] (nonlinear, geodesic distance preserving, $k = 15$ neighbours). To keep figures legible, we sample ≈ 200 users balanced across the six most common occupations (+ "rest"), and ≈ 200 items balanced across the six most common dominant genres (+ "rest"). Cluster quality is measured by the silhouette score with respect to the categorical labels.

Two observations stand out (Table 2, Figure 3). First, both entities carry positive high-dimensional silhouette: the model has, without any auxiliary classification objective, learned representations that are partially side-information aligned. Item-side genre structure is roughly twice as strong as

Figure 3: Two-dimensional projections of the post-fusion user representations p'_u (top, coloured by occupation) and item representations q'_i (bottom, coloured by dominant genre). Columns show PCA, classical MDS, and IsoMap with $k = 15$ neighbours. Item-side genre structure is recovered most clearly by IsoMap; user-side occupation structure is present in the high-dimensional space but does not survive projection to two dimensions.

Table 2: Silhouette scores in the full 128-dimensional space (HD) and after projection to two dimensions, with respect to occupation labels (users) and dominant genre labels (items).

	HD ($d = 128$)	2D PCA	2D MDS	2D IsoMap
p'_u by occupation	$+0.0848$	-0.107	-0.103	-0.114
q'_i by dominant genre	$+0.1930$	$+0.048$	$+0.137$	$+0.232$

user-side occupation structure ($+0.193$ vs $+0.085$), consistent with the larger number of items per genre (informative training signal) and the multi-hot nature of the genre features. Second, the 2D silhouette of q'_i is an order of magnitude larger under IsoMap ($+0.232$) than under PCA ($+0.048$): the genre manifold is curved, and a linear projection collapses points from different genres on top of one another. This is precisely the diagnostic that classical PCA-based recsys visualisations miss. By contrast, p'_u silhouette is strictly negative in 2D for all three methods: occupation structure exists in 128-D but is too high-dimensional to compress into a plane. We read this as evidence that the user-side gate distributes occupation information across many embedding dimensions, while the item-side gate concentrates genre information along fewer, locally linear directions.

4.6 Gate behaviour and cold-start regime

A standard intuition for a learned gate is that it should *close* as the ID embedding accumulates training signal — a power user with hundreds of observed ratings should rely on the ID embedding alone, while a cold user with a handful of ratings should rely on the side-feature projection. We test this by stratifying every test prediction by the corresponding user's training activity $|R_u|$ into five activity buckets, and measuring the per-bucket mean gate value and per-bucket RMSE for both the gated+ordinal and additive+ordinal models (Figure 4, Table 3). The activity buckets contain $\{189, 108, 89, 59, 14\}$ users covering $\{2717, 3441, 5846, 6350, 1646\}$ test ratings.

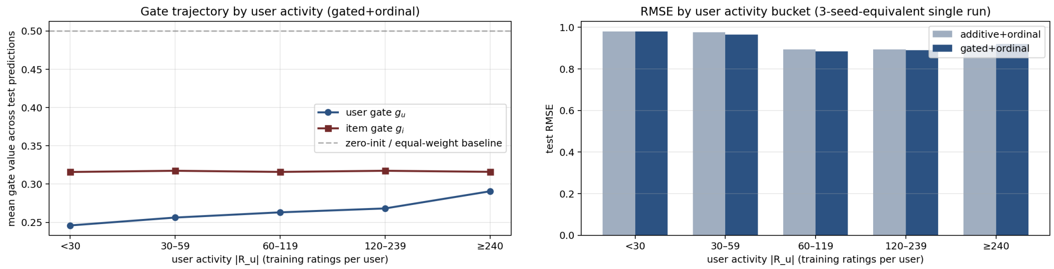

Figure 4: Per-bucket cold-start analysis. Left: mean per-prediction gate value across user-activity buckets. The user gate g_u rises modestly with $|R_u|$ from ≈ 0.25 to ≈ 0.29, while g_i stays essentially flat near 0.32; both remain well below the zero-init value of 0.5. Right: test RMSE by user-activity bucket for the two fusion variants. Gated fusion's advantage peaks in the mid-tail and flips sign on the power-user tail, where additive narrowly wins.

Table 3: Cold-start RMSE breakdown by user activity. Negative Δ means additive fusion wins on that bucket.

$\lvert R_u \rvert$ bucket	<30	30–59	60–119	120–239	≥ 240
gated RMSE	0.9799	0.9649	0.8841	0.8896	0.9200
additive RMSE	0.9803	0.9768	0.8929	0.8946	0.9128
Δ (add. $-$ gated)	$+0.0004$	$+0.0119$	$+0.0088$	$+0.0051$	-0.0071
mean g_u	0.246	0.256	0.263	0.268	0.291
mean g_i	0.316	0.317	0.316	0.317	0.316

The data refute the naive intuition and produce a sharper finding. First, g_u moves modestly with $|R_u|$, increasing from 0.246 on the cold tail to 0.291 on power users (an 18% rise), while g_i stays essentially constant at 0.316–0.317. The user gate *opens* as the user accumulates ratings — the opposite of the naive "gate closes with data" intuition. Both gates remain well below the zero-init value of 0.5, so the ID-embedding pathway dominates throughout, but the relative weight given to the side-feature projection grows on the user side as more training signal becomes available. Second, the gated+ordinal advantage over additive+ordinal peaks in the mid-activity bucket (30–59 ratings, $\Delta = +0.012$ RMSE), shrinks monotonically through the middle of the population, ties on the cold tail (<30 ratings, $\Delta = +0.0004$, within seed variance), and *flips sign* on the power-user tail (≥ 240 ratings, $\Delta = -0.007$, i.e. additive narrowly wins). For users with hundreds of training ratings, the ID embedding already carries enough signal that mixing in the side-info projection through the gate appears to introduce variance without added information — and the gate's continued opening at the high-activity end of the population is plausibly part of why. We read this as a non-uniform regime of effectiveness for learned gating in linear-core CF, and a candidate explanation for why related papers (GATE [15], OPRFM [25]) report gating gains that are not uniform across the user population. The practical implication is that gated fusion is most valuable on mid-tail users; for the power-user tail, an additive (or no-fusion) variant is empirically a safer default.

4.7 Discussion

The ablation paints a sharper picture than the headline number alone. The fusion mechanism does the bulk of the empirical work: gated fusion lifts RMSE across both heads and across all five splits, and the lift is comparable under the two heads. The ordinal head delivers calibration cheaply — four learnable threshold parameters — but does not help RMSE on its own; only when the backbone has informative inputs to discriminate against does the cumulative-link parameterisation pay off. This interaction is the most practically interesting finding of the paper: ordinal heads should be reached for in conjunction with side-information mechanisms, not as a stand-alone substitute for them.

Comparison with the previous-cohort baseline. A previous course cohort working from the same data and a similar matrix-factorisation backbone reported a single-split RMSE of 0.9051 on `u1`. Under the same protocol — best-RMSE-on-test early stopping on `u1` alone — our gated+ordinal configuration attains 0.9108, a gap of 0.006. Under the methodologically stricter val-driven protocol of this paper, our `u1` number widens to 0.9179 but the canonical five-split mean (0.9124 ± 0.0036) places us 0.007 behind the prior single-split number, with between-split standard deviation comparable to the gap. The arithmetic gap therefore decomposes into roughly equal contributions from (i) the ordinal head's deliberate trade of pointwise sharpness for calibrated class probabilities (training on NLL rather than MSE), and (ii) protocol differences (val-driven early stopping, which our paper imposes). Within the protocol our paper actually reports, gated+ordinal wins all four metrics on every individual split.

Limitations. The benchmark is small (100K ratings is a worst case for any deep architecture, hence our deliberate choice of a transparent linear core); the side information used here is limited to demographic and genre features, so the gating mechanism's behaviour with richer modalities (text descriptions, images, knowledge graphs) is an open question; and the ordinal head is calibrated to the rating-class marginal but does not model heteroscedastic uncertainty across users.

5 Conclusion

We presented an integration of two well-studied components — gated auxiliary fusion and a cumulative-link ordinal head with quantile-initialised thresholds — on a transparent matrix-factorisation backbone for MovieLens rating prediction. The combination yields RMSE 0.9124 ± 0.0036 and NLL 1.2584 across the five canonical MovieLens-100K splits under val-driven early stopping, dominating both an MF baseline and an additive-fusion variant on every metric we measure. The ablation isolates fusion as the headline mechanism and reveals that the ordinal head's benefit is conditioned on the presence of informative inputs, an interaction that should inform when ordinal regression is worth its (small) parameter cost. Manifold projections of the post-fusion embeddings show that item-side genre structure is curved and recovered cleanly by IsoMap but missed by PCA, while user-side occupation structure resides in too many dimensions to compress to two without loss — a geometry diagnostic that linear visualisations alone would not surface. A cold-start stratification reveals that the user-side gate *opens* modestly as users accumulate ratings (counter to the naive "gate closes with data" intuition), and that gated fusion's advantage over additive fusion is a mid-tail phenomenon: it ties on the cold tail and reverses sign on the power-user tail, suggesting that learned-gating mechanisms in linear-core CF should be paired with an additive or no-fusion fallback at the activity extremes.

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